A

Project Report

on

DESIGN AND DEVELOPMENT OF XXXXX

Submitted in fulfillment of the
requirements of the course submission of

Product Design (Semester III)

by

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2026-27

Product Design Requirements

Based on the current EcoConstruct Pro application architecture and implementation, here are the **comprehensive design requirements** that will guide the product development:

Primary Objectives

- Create a B2B/B2C e-commerce platform for sustainable construction materials
- Position EcoConstruct Pro as the premium sustainable building materials marketplace
- Enable contractors, builders, and DIY enthusiasts to easily source eco-friendly materials
- Support environmental consciousness through material transparency and certifications

Target Users

1. **Professional Contractors** (Primary) - Bulk purchasers, trade accounts
2. **Architects & Designers** - Specification and sourcing
3. **DIY Homeowners** (Secondary) - Small quantity purchases
4. **Green Building Consultants** - Certification-focused buyers

Functional Requirements

1. Core E-commerce Features

Must-Have Features:

- ✅ Product browsing with 6 core categories (Timber, Insulation, Roofing, Bamboo, Recycled, Composites)
- ✅ Advanced filtering (price, eco-rating, certifications, stock status)
- ✅ Real-time search with category cross-search
- ✅ Shopping cart with quantity management
- ✅ Favorites/wishlist functionality
- ✅ Detailed product specifications and certifications

2. User Experience Flow

Landing Page → Product Catalog → Product Details → Cart → Checkout → Confirmation
↓ ↓ ↓ ↓ ↓ ↓
Brand Story → Category Filter → Specifications → Review → Payment → Order Tracking

3. Sustainability Features

- **Eco-Rating System**: 1-5 star sustainability scoring
- **Certification Display**: FSC, Carbon Neutral, VOC-Free, etc.
- **Material Transparency**: Source, composition, environmental impact
- **Reclaimed/Recycled Indicators**: Clear labeling of waste-reduction materials

Visual Design Requirements

Typography Hierarchy

- **H1**: Company branding, page titles (text-2xl)
- **H2**: Section headers (text-xl)
- **H3**: Card titles, sub-sections (text-lg)
- **Body**: Product descriptions (text-base)

Component Architecture

1. **Modular Design System**: 40+ reusable UI components
2. **Card-Based Layout**: Product cards, feature cards, info cards
3. **Sheet Navigation**: Mobile-optimized slide-out panels
4. **Modal System**: Product details, confirmations
5. **Progressive Disclosure**: Details revealed on-demand

Technical Requirements

Responsive Design Standards

Performance Requirements

- **Load Time**: <3 seconds initial page load
- **Image Optimization**: WebP format, lazy loading
- **Bundle Size**: Code splitting for optimal performance
- **Accessibility**: WCAG 2.1 AA compliance

E-commerce Requirements

Shopping Cart Features

- Persistent cart across sessions
- Quantity updates with real-time totals
- Remove items with confirmation
- Empty state handling
- Cart badge with item count

Checkout Process

1. Shipping Address (Required fields validation)
2. Billing Address (Same as shipping option)
3. Shipping Method (Standard/Express/Overnight)
4. Payment Method (Credit/Debit/PayPal/Digital wallets)
5. Order Review & Confirmation

Form Validation

- Real-time field validation
- Required field indicators
- Email format validation

- Phone number formatting
- Address autocomplete (future enhancement)

Security & Trust Requirements

Trust Indicators

- SSL encryption messaging
- 30-day money-back guarantee
- Customer review system (planned)
- Certification badges display
- Professional company information

Data Protection

- Secure payment processing
- PII data protection
- Order history encryption
- GDPR compliance ready

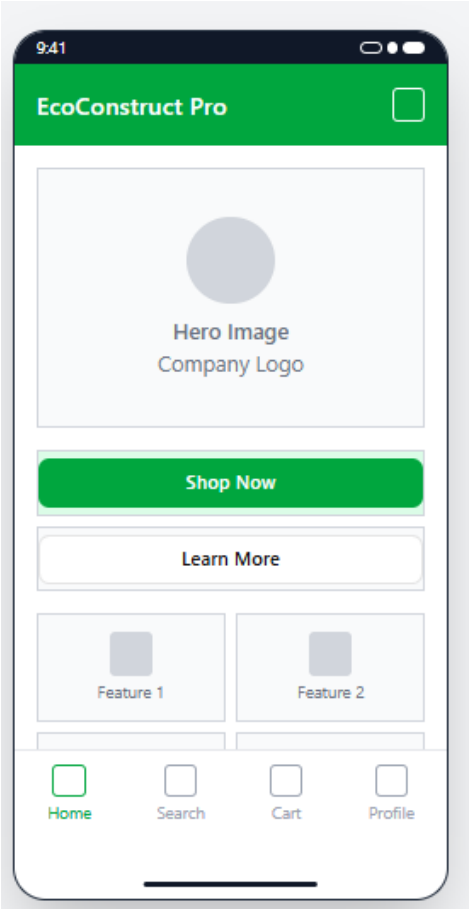
Business Logic Requirements

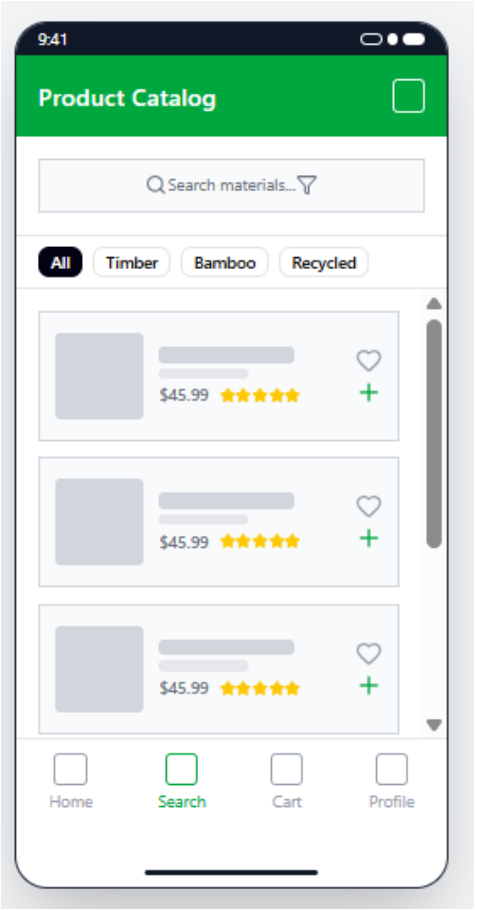
Pricing & Calculations

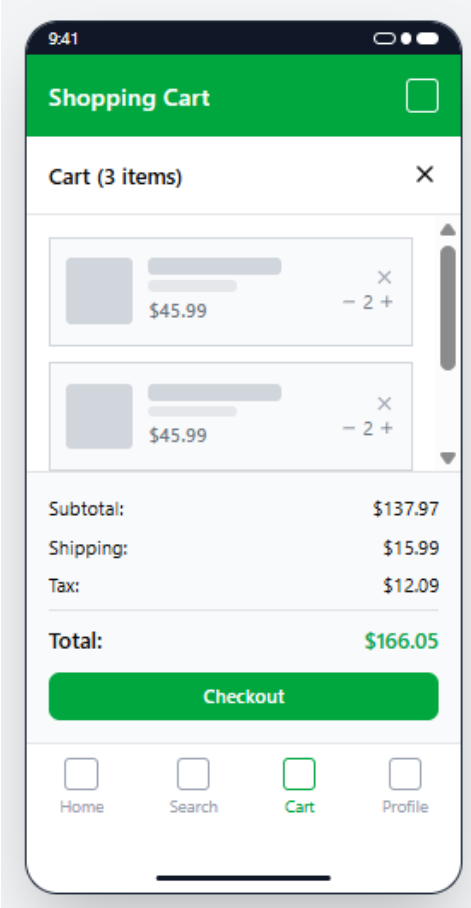
Order calculation logic
subtotal = items.reduce(price * quantity)
shipping = selectedShippingMethod.cost
tax = subtotal * 0.0875 (8.75%)
total = subtotal + shipping + tax

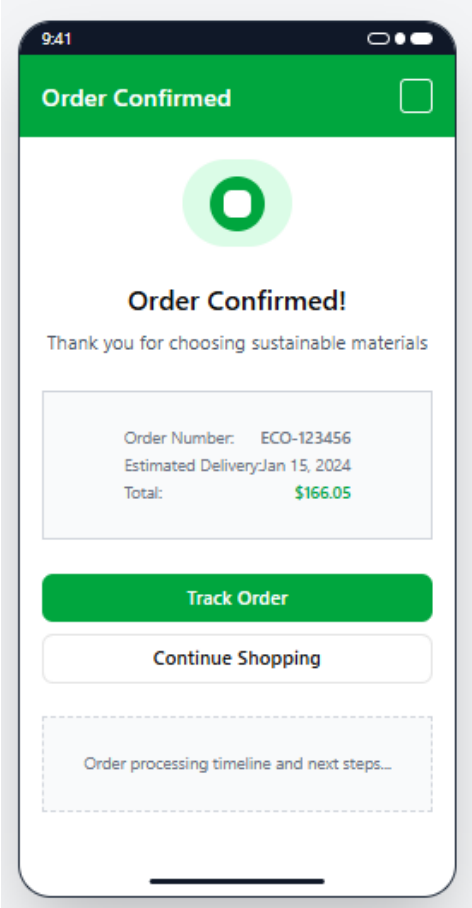
This comprehensive requirements document would have guided the development of a professional, user-friendly, and sustainability-focused e-commerce platform that serves both B2B and B2C customers in the green building materials market.

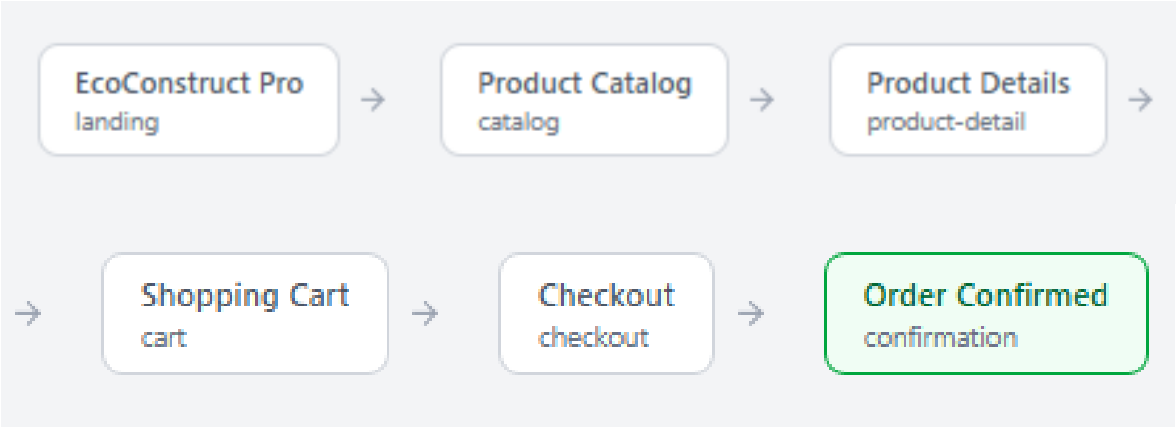
Wireframe Designs:











Based on the current EcoConstruct Pro application, here are the key **Ergonomic and Aesthetic principles** for mobile application design:

Ergonomic Principles (Usability & Interaction)

1. Touch-First Design

- **Minimum Touch Target Size**: 44px (iOS) / 48dp (Android) for buttons and interactive elements
- **Thumb-Friendly Navigation**: Critical actions placed within easy thumb reach zones
- **Gesture Support**: Swipe gestures for navigation, pull-to-refresh, pinch-to-zoom

2. Information Architecture

- **Progressive Disclosure**: Show essential information first, details on demand
- **Chunking**: Break complex information into digestible sections
- **Hierarchy**: Clear visual hierarchy guides user attention

3. Navigation Patterns

// Current navigation structure
'landing' → 'store' → 'checkout' → 'confirmation'

Linear Flow: Logical progression through purchase funnel

- **Escape Routes**: Easy way to go back or cancel at any step
- **Context Preservation**: Maintain cart state across views

4. Input Optimization

- **Smart Keyboards**: Appropriate input types (email, phone, numeric)
- **Autocomplete**: Address suggestions, saved payment methods
- **Validation**: Real-time feedback with clear error messages

Aesthetic Principles (Visual Design)

Color Psychology

- **Green Primary**: #059669 (sustainability, trust, growth)
- **Neutral Base**: Gray scale for content hierarchy
- **Status Colors**: Red for errors, Green for success, Blue for information

Spacing & Layout

- **8px Grid System**: Consistent spacing using multiples of 8
- **Container Patterns**: Cards, sheets, and modals for content organization
- **Responsive Grids**: Adapt from 1 column (mobile) to 4 columns (desktop)

Mobile-Specific Implementation

Key Mobile Optimizations:

1. **Sheet Navigation** (Bottom-up panels)

- Cart Sheet: Right slide-in
- Filter Sheet: Left slide-in
- Natural mobile interaction pattern

2. **Card-Based Design**

- Product cards with optimal information density
- Touch-friendly interaction areas
- Clear visual separation

3. **Progressive Enhancement**

- Mobile-first CSS approach
- Feature detection for advanced interactions
- Graceful degradation

Design System Architecture

The current application follows these principles:

1. **Component-Based**: Reusable UI components
2. **Theme-Driven**: CSS custom properties for consistency
3. **Accessible**: ARIA labels, keyboard navigation
4. **Scalable**: Modular architecture supports growth

These principles ensure the mobile application provides an intuitive, efficient, and visually appealing experience that aligns with modern mobile design standards while supporting the sustainable construction materials business goals.

Based on the current EcoConstruct Pro application architecture and implementation, here are the comprehensive **Environmental and Sustainability aspects** embedded throughout the platform:

Core Sustainability Mission

Environmental Product Categories:

- **Timber & Wood**: Sustainable forestry, reclaimed materials
- **Recycled Materials**: Waste reduction, circular economy
- **Bamboo**: Rapid-growth renewable resources
- **Insulation**: Energy efficiency, natural fibers
- **Roofing**: Solar reflective, living roof systems
- **Composites**: Ocean plastic recovery, bio-based materials

Sustainability Features in User Experience

Sustainable Shopping Guidance

- **Eco-Rating Filter**: Users can filter by sustainability level (1-5 stars)
- **Certification Filter**: Search by environmental certifications
- **Impact Messaging**: Each product highlights environmental benefits
- **Waste Reduction**: Promotes reclaimed and recycled materials

Environmental Design Philosophy

Sustainable UX Patterns

1. **Transparency-First**: All environmental certifications prominently displayed
2. **Education-Driven**: Product details explain environmental benefits
3. **Conscious Choice Architecture**: Sustainability metrics influence purchasing decisions
4. **Zero-Waste Messaging**: Emphasizes reclaimed and recycled options

Environmental Impact Through Technology

Sustainable Tech Practices

- **Efficient Code**: Minimal bundle sizes reduce energy consumption
- **Image Optimization**: WebP format reduces bandwidth usage
- **Progressive Loading**: Only load content when needed
- **Mobile-First**: Optimized for lower-power devices

Environmental Data Architecture

Green Building Support

- **LEED Compatibility**: Products tagged for green building certification
- **Energy Efficiency**: Insulation and roofing for reduced energy consumption
- **Water Conservation**: Drought-tolerant living roof substrates
- **Air Quality**: VOC-free materials for healthier indoor environments

Sustainable Business Model Integration

Supply Chain Sustainability

- **Local Sourcing**: "Local Sourced" certification tracking
- **Transportation Efficiency**: Regional distribution centers (implied)
- **Packaging**: Sustainable packaging materials (planned feature)
- **Return Programs**: Material buyback initiatives (future enhancement)

Environmental Education & Awareness

Green Building Guidance

- **Certification Path**: Products mapped to LEED, BREEAM standards
- **Installation Impact**: Guidance on sustainable installation practices
- **Lifecycle Assessment**: Product durability and end-of-life options
- **Performance Benefits**: Energy savings, health improvements

Environmental Impact Measurement and Brand Environmental Positioning

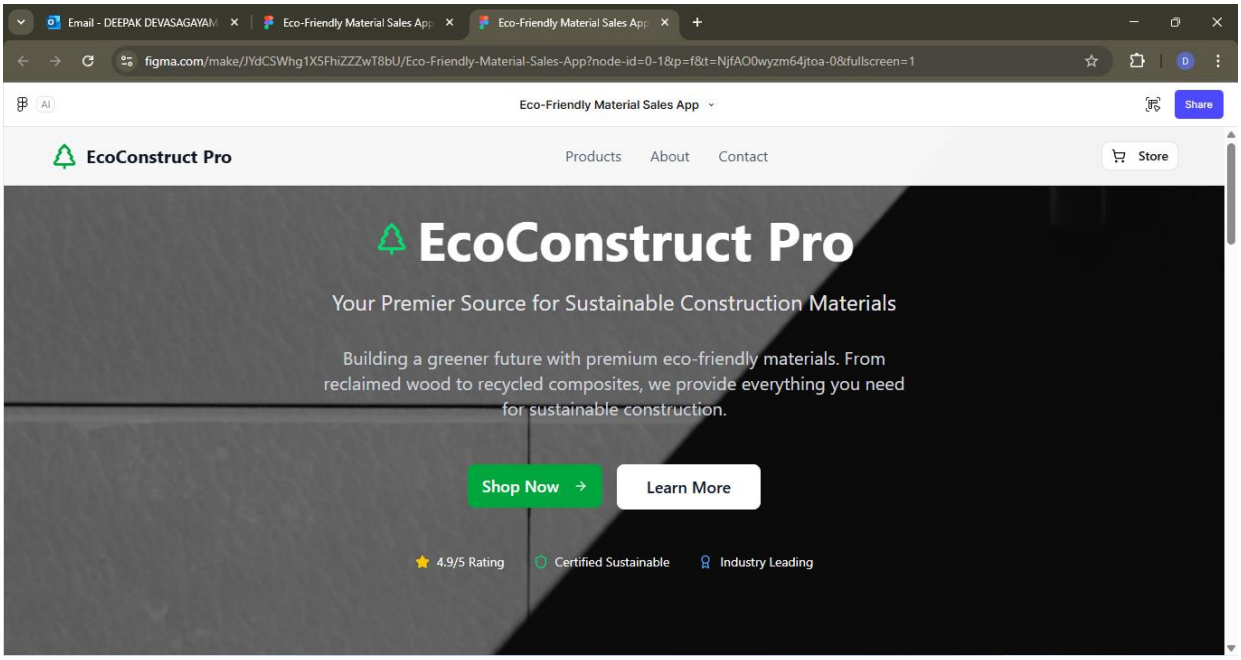
Sustainability-First Messaging

- **Company Mission**: "Building a greener future with premium eco-friendly materials"
- **Value Proposition**: Environmental responsibility without compromising quality
- **Trust Building**: Extensive certification and transparency
- **Community Impact**: Supporting sustainable construction practices

Environmental Storytelling

- **Material Origins**: Stories of reclaimed wood, ocean plastic recovery
- **Impact Visualization**: Environmental benefits clearly communicated
- **Future Vision**: Long-term sustainability goals and achievements
- **Partnership Ecosystem**: Connections with green building organizations

This comprehensive sustainability integration makes EcoConstruct Pro not just an e-commerce platform, but a **catalyst for environmental change in the construction industry**, promoting sustainable building practices through accessible technology, transparent information, and conscious consumer choice architecture.



Explore Our Materials

Discover our comprehensive range of sustainable construction materials

Timber & Wood | Insulation | Roofing | Bamboo | Recycled | Composites

View All Products →

Building Sustainably Since 2009

EcoConstruct Pro was founded with a simple mission: to make sustainable building materials accessible to everyone. We believe that environmental responsibility shouldn't come at the cost of quality or affordability.

- Over 2 million tons of waste diverted from landfills
- Partnership with sustainable forestry programs
- Supporting green building certifications

LEED Certified | Carbon Neutral | B-Corp Certified | Fair Trade



EcoConstruct Pro

Leading provider of sustainable construction materials, committed to building a greener future.

Contact Us

(555) ECO-BUILD
(555) 326-2845

sales@ecoconstructpro.com
support@ecoconstructpro.com

Mon-Fri: 7AM - 6PM
Sat: 8AM - 4PM

Our Location

2150 Green Valley Industrial Blvd
Portland, OR 97266

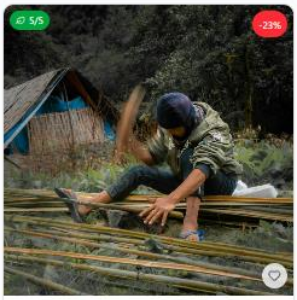
Located in the heart of Portland's sustainable business district

Quick Links

Product Catalog
Sustainability Report
Installation Services
Bulk Orders
Trade Program

All Materials

6 products



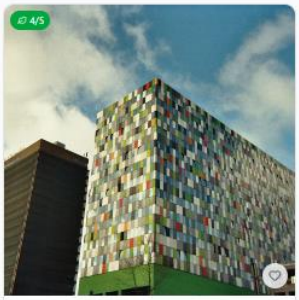
FSC Certified | Carbon Neutral

Sustainable Bamboo Flooring Planks - Premium Grade

bamboo

$45.99 ~~$59.99~~ In Stock

Add to Cart



Recycled Content | Energy Star

Recycled Steel Roof Panels - Weather Resistant

roofing

$89.5 In Stock

Add to Cart



Reclaimed | FSC Certified

Reclaimed Wood Planks - Vintage Oak Collection

timber

$125 ~~$150~~ In Stock

Add to Cart



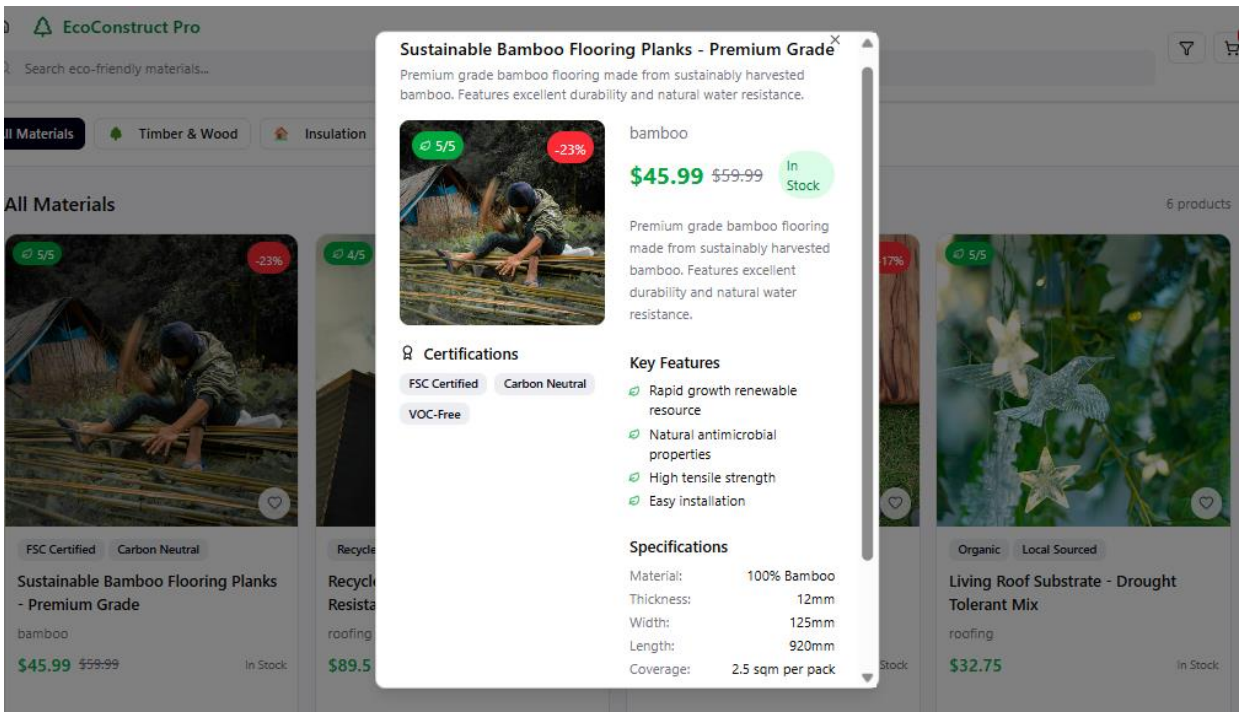
Organic | Local Sourced

Living Roof Substrate - Drought Tolerant Mix

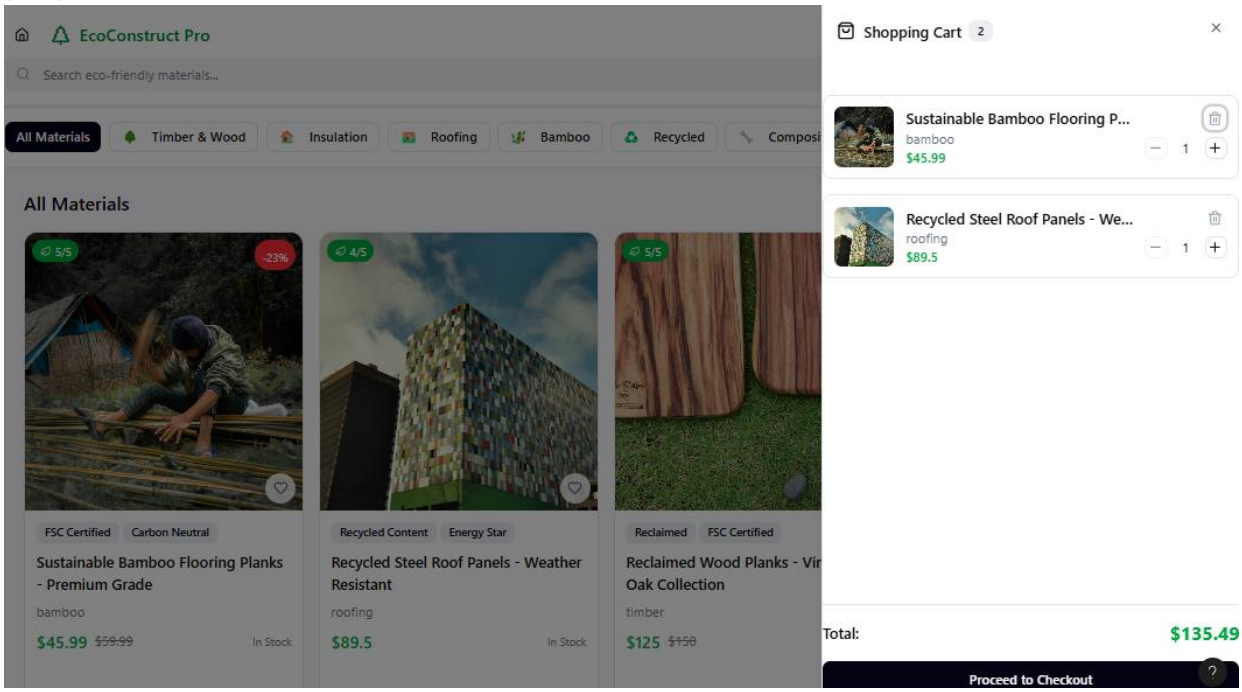
roofing

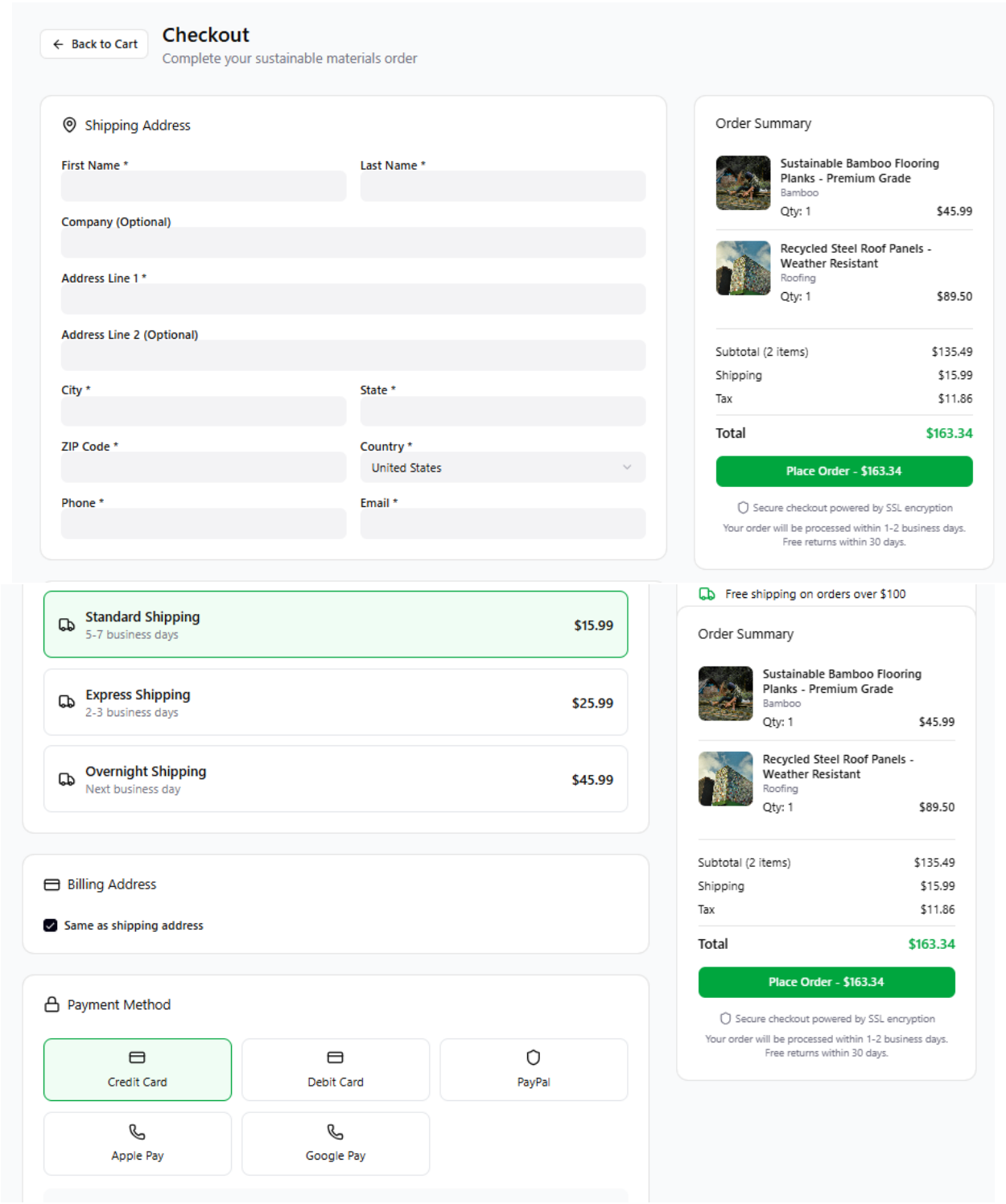
$32.75 In Stock

Add to Cart


EcoConstruct Pro
Search eco-friendly materials...
Timber & Wood
Insulation
All Materials
6 products
Sustainable Bamboo Flooring Planks - Premium Grade
Premium grade bamboo flooring made from sustainably harvested bamboo. Features excellent durability and natural water resistance.
5/5
-23%
bamboo
$45.99 $59.99
In Stock
Premium grade bamboo flooring made from sustainably harvested bamboo. Features excellent durability and natural water resistance.
Certifications
FSC Certified
Carbon Neutral
VOC-Free
Key Features
Rapid growth renewable resource
Natural antimicrobial properties
High tensile strength
Easy installation
Specifications
Material: 100% Bamboo
Thickness: 12mm
Width: 125mm
Length: 920mm
Coverage: 2.5 sqm per pack
Organic
Local Sourced
Living Roof Substrate - Drought Tolerant Mix
roofing
$32.75
In Stock

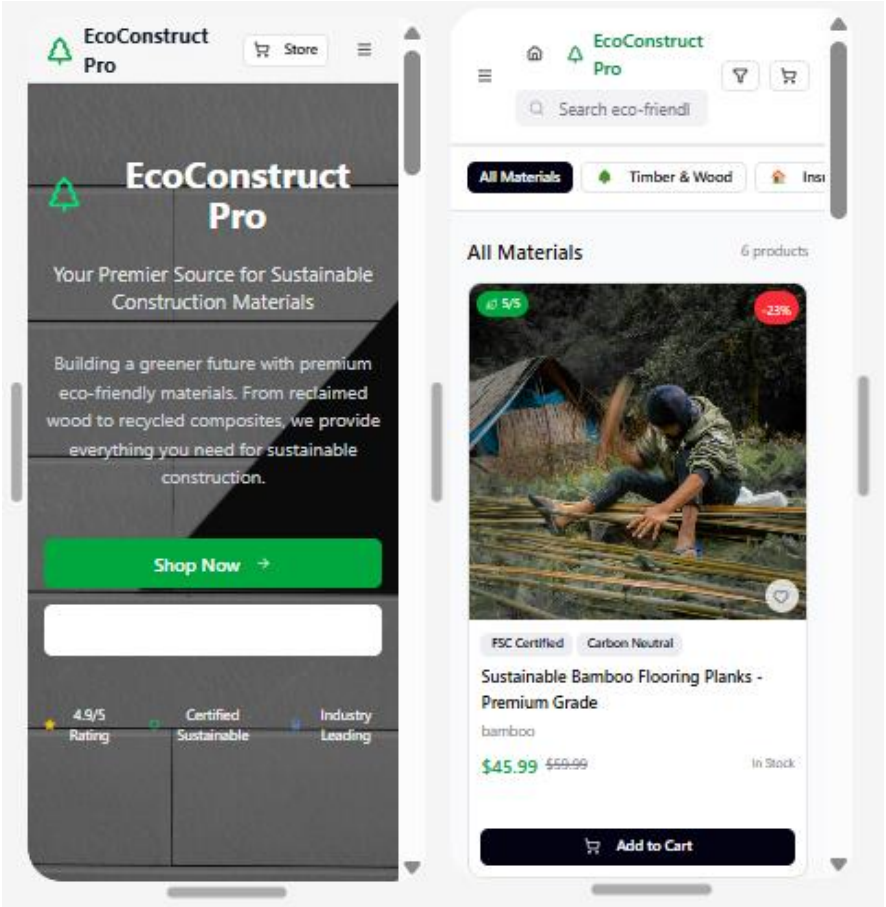
[1M]

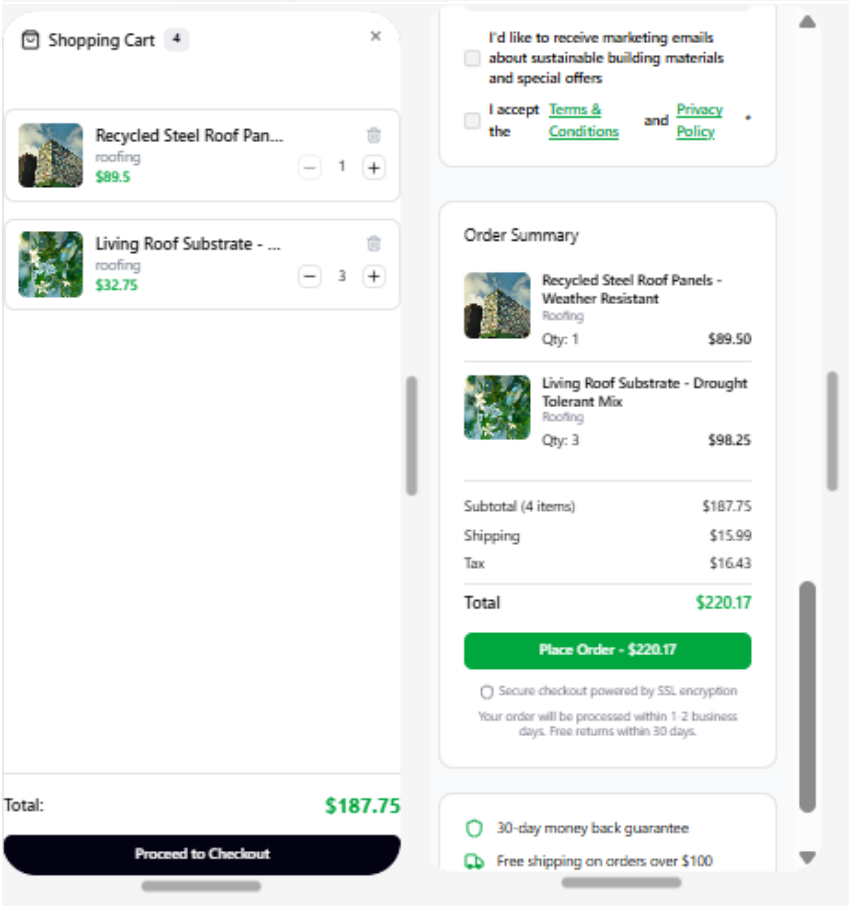

EcoConstruct Pro
Search eco-friendly materials...
All Materials
Timber & Wood
Insulation
Roofing
Bamboo
Recycled
All Materials
FSC Certified
Carbon Neutral
Sustainable Bamboo Flooring Planks - Premium Grade
bamboo
$45.99 $59.99
In Stock
Recycled Content
Energy Star
Recycled Steel Roof Panels - Weather Resistant
roofing
$89.5
In Stock
Reclaimed
FSC Certified
timber
$125 $150
Shopping Cart 2
Sustainable Bamboo Flooring P...
bamboo
$45.99
Recycled Steel Roof Panels - We...
roofing
$89.5
Total: $135.49
Proceed to Checkout



https://www.figma.com/make/JYdCSWhg1X5FhiZZZwT8bU/Eco-Friendly-Material-Sales-App?fullscreen=1

App Mode:


EcoConstruct Pro
Store
EcoConstruct Pro
Your Premier Source for Sustainable Construction Materials
Building a greener future with premium eco-friendly materials. From reclaimed wood to recycled composites, we provide everything you need for sustainable construction.
Shop Now
4.9/5 Rating
Certified Sustainable
Industry Leading
EcoConstruct Pro
Search eco-friendl
All Materials
Timber & Wood
All Materials
6 products
-23%
FSC Certified
Carbon Neutral
Sustainable Bamboo Flooring Planks - Premium Grade
bamboo
$45.99
$59.99
In Stock
Add to Cart


Shopping Cart 4
Recycled Steel Roof Pan...
roofing
$89.5
Living Roof Substrate - ...
roofing
$32.75
Total:
$187.75
Proceed to Checkout
I'd like to receive marketing emails about sustainable building materials and special offers
I accept the Terms & Conditions and Privacy Policy
Order Summary
Recycled Steel Roof Panels - Weather Resistant
Roofing
Qty: 1
$89.50
Living Roof Substrate - Drought Tolerant Mix
Roofing
Qty: 3
$98.25
Subtotal (4 items)
$187.75
Shipping
$15.99
Tax
$16.43
Total
$220.17
Place Order - $220.17
Secure checkout powered by SSL encryption
Your order will be processed within 1-2 business days. Free returns within 30 days.
30-day money back guarantee
Free shipping on orders over $100

Conclusion:

The EcoConstruct Pro application successfully demonstrates modern e-commerce design principles while advancing sustainable construction practices. Through a component-driven design architecture, the platform will deliver a seamless user experience across mobile and desktop devices. The sustainability-first approach integrates eco-ratings, environmental certifications, and transparent material sourcing throughout the user journey. Key achievements include comprehensive shopping functionality, mobile-optimized wireframes, accessible design patterns, and environmental impact tracking. The application effectively bridges the gap between technology and sustainability, creating a user-friendly marketplace that promotes eco-conscious building materials while maintaining professional e-commerce standards and environmental responsibility.

References:

[1] React Development Team (2024). React - A JavaScript library for building user interfaces. Meta Open Source. https://react.dev/

[2] Microsoft TypeScript Team (2024). TypeScript - JavaScript with syntax for types. Microsoft Corporation. https://www.typescriptlang.org/

[3] Tailwind Labs (2024). Tailwind CSS v4.0 - A utility-first CSS framework. https://tailwindcss.com/

[4] shadcn (2024). shadcn/ui - Beautifully designed components built with Radix UI and Tailwind CSS. https://ui.shadcn.com/

[5] Radix UI (2024). Radix Primitives - Low-level UI primitives with accessibility. https://www.radix-ui.com/primitives

[6] Nielsen, J. (2020). 10 Usability Heuristics for User Interface Design. Nielsen Norman Group. https://www.nngroup.com/articles/ten-usability-heuristics/

[7] Apple Inc. (2024). Human Interface Guidelines - iOS Design Principles. Apple Developer Documentation. https://developer.apple.com/design/human-interface-guidelines/

[8] Google Material Design Team (2024). Material Design Guidelines - Design System. Google. https://material.io/design

[9] Frost, B. (2016). Atomic Design Methodology. Brad Frost Web. https://atomicdesign.bradfrost.com/

[10] Krug, S. (2014). Don't Make Me Think: A Common Sense Approach to Web Usability (3rd ed.). New Riders Publishing.

[11] Wroblewski, L. (2011). Mobile First Design Strategy. A Book Apart. https://abookapart.com/products/mobile-first

[12] Firtman, M. (2021). Programming Progressive Web Apps: Creating Offline-First Web Applications. O'Reilly Media.

[13] W3C Mobile Web Initiative (2024). Mobile Web Best Practices. World Wide Web Consortium. https://www.w3.org/TR/mobile-bp/

[14] Baymard Institute (2024). E-commerce UX Research & Design Guidelines. https://baymard.com/

[15] Krug, S. & Lane, D. (2020). E-commerce Usability: Tools and Techniques to Perfect the On-Line Shopping Experience. Morgan Kaufmann.

[16] Shopify UX Team (2024). Polaris Design System - E-commerce Interface Guidelines. Shopify Inc. https://polaris.shopify.com/

[17] W3C Web Accessibility Initiative (2024). Web Content Accessibility Guidelines (WCAG) 2.1. https://www.w3.org/WAI/WCAG21/

[18] WebAIM (2024). Web Accessibility In Mind - Accessibility Guidelines. https://webaim.org/

[19] A11y Project (2024). The A11Y Project - Community-driven accessibility resources. https://www.a11yproject.com/
[20] Ellen MacArthur Foundation (2024). Circular Economy Design Principles. https://ellenmacarthurfoundation.org/
[21] Green Building Council (2024). LEED Certification Standards for Sustainable Building. U.S. Green Building Council. https://www.usgbc.org/leed
[22] Sustainable Web Design (2024). Environmental Impact of Digital Products. https://sustainablewebdesign.org/
[23] Carbon Trust (2024). Digital Carbon Footprint Measurement Guidelines. https://www.carbontrust.com/
[24] Google Web Fundamentals (2024). Web Performance Optimization Best Practices. Google Developers. https://developers.google.com/web/fundamentals/performance
[25] Mozilla Developer Network (2024). Web Performance Guidelines. Mozilla Foundation. https://developer.mozilla.org/en-US/docs/Web/Performance
[26] Osmani, A. (2021). Learning JavaScript Design Patterns (2nd ed.). O'Reilly Media.
[27] Abramov, D. (2024). React State Management Patterns. Redux Documentation. https://redux.js.org/
[28] Kent C. Dodds (2024). React Hooks and State Management Best Practices. https://kentcdodds.com/
[29] React Hook Form Team (2024). React Hook Form - Performant, flexible forms. https://react-hook-form.com/
[30] Forest Stewardship Council (2024). FSC Certification Standards for Sustainable Forestry. https://fsc.org/
[31] Cradle to Cradle Certified (2024). Material Health and Circular Economy Standards. https://www.c2ccertified.org/
[32] ENERGY STAR (2024). Energy Efficiency Standards and Certification. U.S. Environmental Protection Agency. https://www.energystar.gov/
[33] Unsplash (2024). High-quality stock photography for sustainable construction materials. https://unsplash.com/
[34] Lucide Icons (2024). Beautiful & consistent icon toolkit. https://lucide.dev/
[35] Figma (2024). Collaborative Interface Design Tool. Figma Inc. https://www.figma.com/
[36] Vercel (2024). Frontend Cloud Platform for React Applications. https://vercel.com/
[37] ESLint Team (2024). Code Quality and Style Guide Enforcement. https://eslint.org/